China Spotlight
The Great Demographic Challenge

January 25, 2022

Gene Ma, Head of China Research, gma@iif.com, +1 202 857 3305, @genemaIIF
Phoebe Feng, Senior Research Analyst, pfeng@iif.com, +86 5836 9080, @phoebefeng_xy
Yijia Dai, Research Assistant, ydai@iif.com, +1 202 378 3654

- China's birth rate fell to a 72-year record low.
- The rapidly aging population is having a profound impact on savings, investment, and economic growth.
- Education, urbanization, automation, and later retirement may cushion this demographic shock.
- It remains to be seen how China's fertility rate will respond to Beijing's reversal of its population policies.

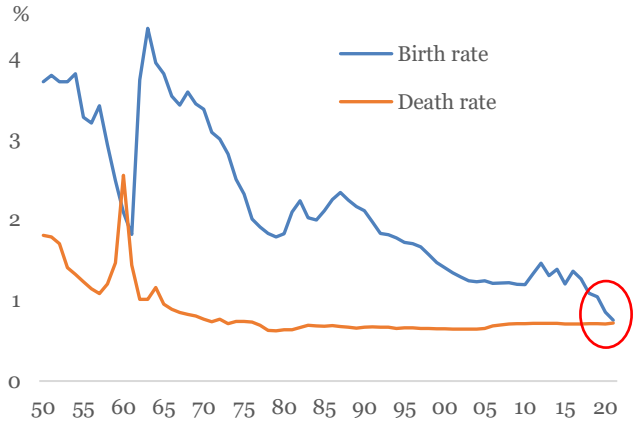
China's latest census, released in 2021, revealed an alarming deceleration in population growth. The forecast for the future population trajectory is also bleak. An aging population is a powerful force that determines China's savings, investment, consumption, housing, inflation, economic growth, and a host of social issues. Without a forceful policy response, China will rapidly acquire the world's largest aging population and reach there at the most rapid speed before getting rich. Some policy adjustments were made last July, though how China's fertility rate will respond to these more supportive policies is unclear.

The alarming decline of population growth

China's birth rate in 2021 declined to 0.75%, a record low across its 72-year data. The birth rate bumped up in 2012-2016, thanks to the relaxation of the one-child policy and baby-boom echo, but plummeted again in the past three years (Chart 1). Simultaneously, China's death rate is gradually climbing due to its aging population. As birth and death rates converged, the natural population growth rate (birth minus death) dropped to 0.034% (Chart 2). China's population stopped growing.

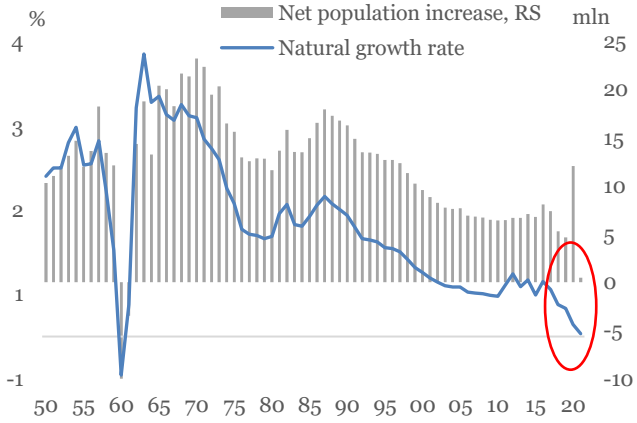
China's fertility rate (the average number of children per woman aged 15-49) rapidly declined from 6.1 in the 1960s to the natural replacement rate (the rate needed to keep the population steady) of 2.1 in 1991 and 1.7 in 2020, according to UNWPP (United Nations World Population Prospects) estimates. China's own national consensus indicated that the fertility rate improved from 1.18 in 2010 to 1.3 in 2020 following policy relaxation. However, this is still among the world's lowest and considerably below the natural replacement rate. For comparison, the average fertility rate is currently 1.6 for developed economies and 2.8 for developing economies (not including China) (Chart 3). Both China's birth rate and fertility rate are now similar to Japan's, although having 1/4 of Japan's per capita GDP.

Chart 1. China's birth rate hit a new record low in 2021



Source: Haver, IIF

Chart 2. The population increase was near zero in 2021



Source: Haver, IIF

China's working-age population (age 15-64) peaked in 2013. It has declined by 38 million in the past seven years, dropping 21.4 million in 2020 alone. The elderly population (65+) increased from 6.4% of the total population in the 1990s to 13.5% in 2020 (Chart 4). The old-age dependency ratio has doubled over the past 20 years to 20%. The population of ethnic minorities increased from 5.8% in 1964 to 8.9% in 2020, as the one-child policy was not enforced on minorities.

Bleak forecast for the future population

The United Nations World Population Prospects tended to be sanguine about China's population trajectory. China's own official forecasts, especially those made during the one-child policy era, were even more off the mark. This section adopts the lower-case forecast by WPP and research conducted by a team of Chinese domestic demographers (link).

It is forecast that China's elderly population (65+) will increase from 13.5% of the total population to about 35% by 2056. (This ratio is currently 26% for Japan.) China's working-age population (15-64) is already in decline and is likely to decrease by an average of 10 million people a year over the remainder of the century (Chart 5). By the end of this century, China's working-age population will have dropped to 580 mln, approximately 57% of today's size.

Even based on the usually sanguine forecast by the UNWPP, China's population will likely peak at 1.46 billion, be surpassed by India around 2030, and fall to below 1.1 billion by 2100. Since China's population is projected to peak approximately 40 years ahead of the global average, its share of the global population will drop from 19% to 8% by 2100.

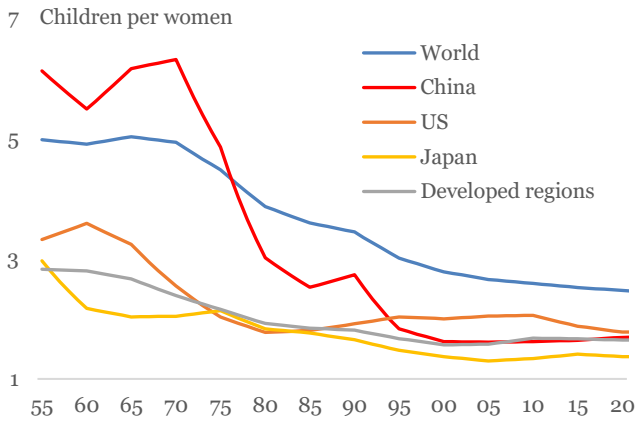
An aging economy

China's economy benefited from the demographic dividend over the past three decades, as the baby boom in the 1960s and later the one-child policy lowered China's dependency ratio from 62.6 in 1982 to 34.2 in 2010 (Chart 6). The rural-to-urban migration further boosted labor productivity. Unfortunately, China's demographic sweet spot has come to an end, with the dependency ratio rising to 45.9 in 2020 and another 400 mln baby-boomers retiring in the next five years. Moreover, the average age of China's migrant workers rose to 41 in 2020 from 33 just 10 years ago.

An aging population means lower savings, investment, consumption, and economic growth. China's working-age population as a share of the total population peaked in 2010 at 74.5%. In the ensuing decade, China's household savings rate declined from its peak of 42.1% to 34.8% (Exhibit 7), the growth of real fixed asset investments slowed from 20.9% to 2.5%, and real GDP growth fell from 10.6% to 2.2%. Population aging will also profoundly impact China's inflation dynamics and capital market development.

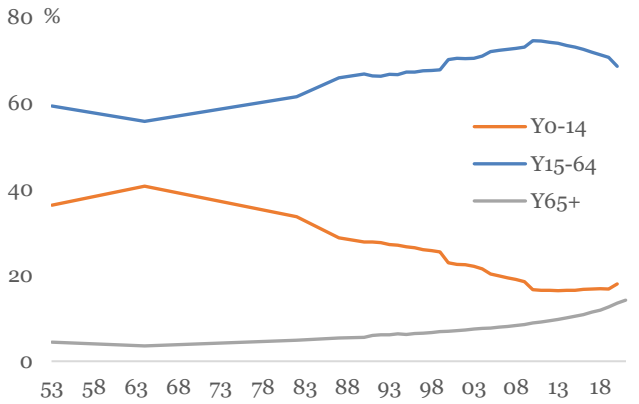
As China's marriage-age population peaked in 2011, the number of newly married people dropped rapidly from 27 mln in 2013 to only 16.2 mln in 2020 (Chart 8). Covid-19 may distort the marriage numbers for 2020. Slower family

Chart 3. The declining fertility rate



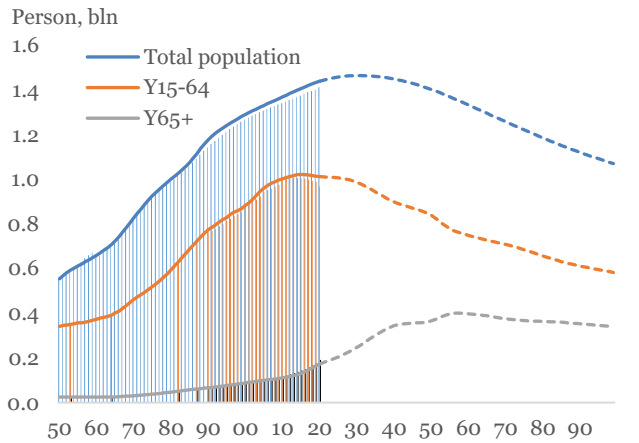
Source: Haver, IIF

Chart 4. The working-age population is declining



Source: Haver, IIF

Chart 5. Population and forecast



Source: Haver, UNWPP
Bars are CNBS data; Lines are UNWPP data.

iif.com © Copyright 2022. The Institute of International Finance, Inc. All rights reserved.

formation is another reason for China's housing boom reaching its peak. Moreover, the average family size dropped from 4.4 in 1982 to 2.6 in 2020. Chinese society has shifted from "four generations under one roof" to one full of childless families.

Mitigating the demographic crisis

Fortunately, education, urbanization, automation, and later retirement can help cushion China's demographic shock. China's illiteracy rate declined from 22.8% in 1982 to 2.7% in 2020. The college-educated population increased to 15.5% in 2020 from 8.9% in 2010 and 0.6% in 1982. The average schooling of the Y16-59 population increased to 10.8 years in 2020 from 9.7 years in 2010.

China's life expectancy increased from 67.8 in 1982 to 77.3 in 2019, only 1.5 years less than the U.S. (78.8). However, China's average retirement age at 54 is 13 years earlier than in the U.S., which is approximately 67 for those born after 1960. China's 14th Five Year Program planned to gradually postpone the legal retirement age, which is currently extremely low at 50 for female blue-collar workers, 55 for female office workers, and 60 for men.

There is still room for further urbanization in China since its urbanization rate (64%) is still below developed economies' (80%). China's National Population Development Prospect (2016—2030) expects a further increase in education levels and urbanization rate (Table 1).

Table 1. Population Development Prospect 2016—2030

Main Indicators		2015	2020	2030
Total Population	bln	1.375	1.42	1.45
Total Fertility Rate		1.5-1.6	1.8	1.8
Sex Ratio at Birth	M/FM	113.5	≤112	107
Average Life Expectancy	Age	76.3	77.3	79
Labor's Average schooling	Year	10.23	10.8	11.8
Urbanization Rate	%	56.1	60	70

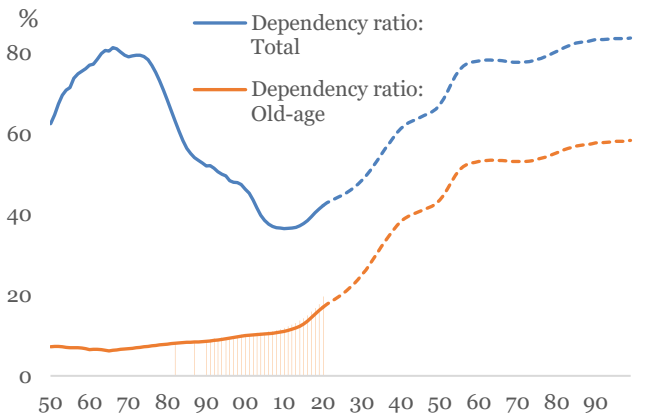
Source: National Bureau of Statistics

Moreover, automation, artificial intelligence, and the growing share of the digital and knowledge economy will mean China's economy depends more on education instead of the size of its labor force. China's annual output of industrial robots in 2020 was 10 times that of five years ago, and investments in technology grew at double digits during 2013-19.

Policy adjustments on the way

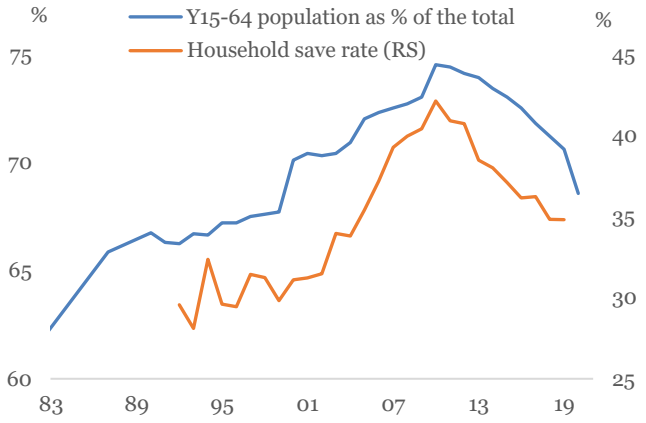
Tackling the aging problem featured prominently in the 14th Five Year Program (2021-25). Last July, the new policy support for childbearing covered maternity benefits, education, tax incentives, housing, and women's employment to "secure stable long-term population development by 2035". It remains to be seen how China's fertility rate will respond to these policy adjustments; however, without them, China will soon have the world's oldest population and will arrive there the quickest.

Chart 6. Dependency ratio and forecast



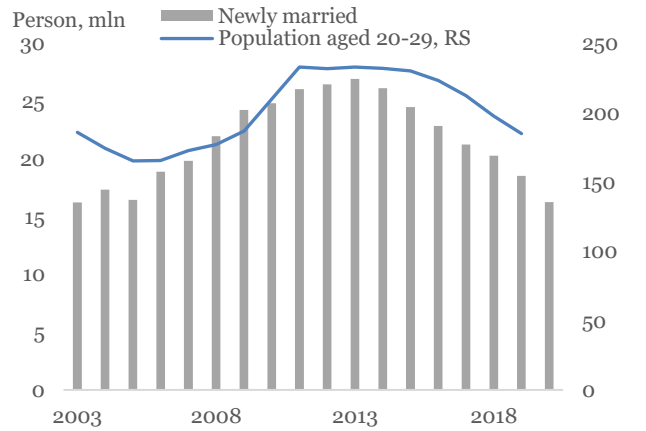
Source: Haver, UNWPP
Bars are CNBS data; Lines are UNWPP data.

Chart 7. Saving rate declines as the population get older



Source: Haver, IIF

Chart 8. Fewer are getting married



Source: Have, IIF

iif.com © Copyright 2022. The Institute of International Finance, Inc. All rights reserved.